early allows for just-in-time ordering of spares and maintenance scheduling, optimizing inventory and labor.

### 2.1.2 Establish measurable performance indicators and baselines

To illustrate how digital twins can predict and prevent unplanned downtime in a manufacturing plant, we can examine both leading indicators (early warning signals) and lagging indicators (business impact metrics) that work together to provide comprehensive operational insight as shown in table 2.1.

**Table 2.1 Leading and lagging indicators defined for predictive maintenance of centrifugal pumps.**

| Indicator type | Description | Baseline | Target |
|---|---|---|---|
| **Leading** (Early warning) | Fluctuations in pump vibration and temperature (precursors to bearing failure) | Vibration: ±0.5 $mm/s^2$. Temp: ±1°C weekly | Fluctuation alerts at >2.0 $mm/s^2$ and >3°C weekly |
| **Lagging** (Business outcome) | Reduction in unplanned downtime and unnecessary replacements | 12 hours/month downtime. 40% unnecessary replacement | 3 hours/month downtime (75% reduction, $1.62 million saving) 60% reduction in unnecessary replacements ($29K saving) |

### 2.1.3 Map decision-making improvements

Next, you should articulate which specific decisions will be enhanced and how actions taken based on those decisions will deliver business value. Document current decision-making processes, identify information gaps, and define how the digital twin provides better, faster, or more confident decisions, as shown in table 2.2 for the example of pump maintenance.